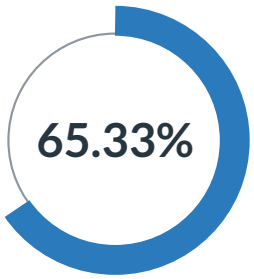
Results

Carter Feng — 1st Attempt



65.333
Out of 100 points

37:35:23
Time for this attempt

This assessment is locked.

Attempt History

Results	Points	Score	(First score is kept)
Attempt 1	65.333 of 100	65.33%	(First score)

Your Answers:

1 — 10 / 10 points — Matching

Create a red-black tree using the following values:37, 42, 47, 21, 4

What are the following nodes values when the 2nd outside-outside ro

Red Parent Node	✓ 21

Red Child (or X) Node	✓ 18
What type of rotation is it?	✓ left - left
Black Grandparent Node	✓ 37
Black Sibling Node	✓ null

2 10 / 10 points Fill in the Blank

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21, 4

What is the black height of the root?

✓ 3

3 10 / 10 points Matching

Create a red-black tree using the following values:37, 42, 47, 21, 4

What are the following nodes values when the 3rd outside-outside ro

Red Child (or X) Node	✓ 6

Black Sibling Node	✓ null
Red Parent Node	✓ 9
What type of rotation is it?	✓ left - left
Black Grandparent Node	✓ 18

4 2 / 10 points Fill in the Blank

`SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43`

Create a red-black tree using the following values:37, 42, 47, 21, 46, 18, 30, 33, 9, 6, 7, 45, 3, 29, 43.What is the preorder traversal of all the red nodes in the red-black tree? Enter null when nothing is present. Only enter value, not color.

✗ null Correct Answer: **3**

✗ 9 Correct Answer: **7**

✗ n Correct Answer: **33**

✗ 3 Correct Answer: **29**

✗ 7 Correct Answer: **46**

✗ null Correct Answer: **43**

✗ 42 Correct Answer: **null**

✓ null

✗ 29 Correct Answer: **null**

✗ 33 Correct Answer: **null**

✓ null

✗ 45 Correct Answer: **null**

✗ 47 Correct Answer: **null**

✗ n Correct Answer: **null**

✓ null

5 2.667 / 10 points Fill in the Blank

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21, 46, 18, 30, 33, 9, 6, 7, 45, 3, 29, 43.What is the post order traversal of all the black nodes in the red-black tree? Enter null when nothing is present. Only enter value, not color.

✗ null
Correct Answer: **6**

✗ null
Correct Answer: **18**

✗ 6
Correct Answer: **9**

✗ 18
Correct Answer: **30**

✗ null
Correct Answer: **37**

✗ null
Correct Answer: **45**

✗ null
Correct Answer: **47**

✗ 30
Correct Answer: **42**

✗ null
Correct Answer: **21**

✓ null

✗ 46
Correct Answer: **null**

✓ null

✗ 21
Correct Answer: **null**

✓ null

✓ null

6 10 / 10 points Fill in the Blank

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21,

What is the value of the root?

✓ 21

7 6.667 / 10 points Fill in the Blank

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21, 46, 18, 30, 33, 9, 6, 7, 45, 3, 29, 43.What is the level order traversal of all the black nodes in the red-black tree? Enter null when nothing is present. Only enter value, not color.

✓ 21 ✗ null

Correct Answer: **9**

✗ null ✓ 6 ✓ 18 ✓ 30 ✗ 46

Correct Answer: **42** Correct Answer: **37**

✗ null ✗ null ✓ null ✓ null ✓ null

Correct Answer: **45** Correct Answer: **47**

✓ null ✓ null ✓ null

8 0 / 10 points Matching

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21,

What are the following nodes values when the 5th color swap (siblin

Red Left Child Node — ✗ 45

Correct Answer: **30**

Black Parent Node — ✗ 46

Correct Answer: **33**

Red Right Child Node — ✗ 47

Correct Answer: **37**

9 10 / 10 points Fill in the Blank

SET 1: 37 42 47 21 46 18 30 33 9 6 7 45 3 29 43

Create a red-black tree using the following values:37, 42, 47, 21,

How many null nodes are on level 3 of the tree?

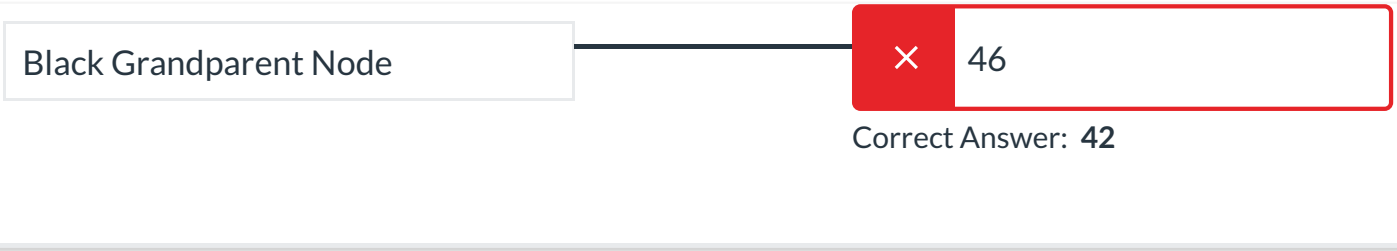
✓ 2

10 4 / 10 points Matching

Create a red-black tree using the following values:37, 42, 47, 21,

What are the following nodes values when the 4th outside-outside ro

What type of rotation is it?	✓ left - left
Red Parent Node	✗ 45
	Correct Answer: **21**
Black Sibling Node	✓ 47
Red Child (or X) Node	✗ 43
	Correct Answer: **9**



Correct Answer: **42**